PriorSplat: Compositional Regression in End-to-End 3D Priors for Per-Scene Gaussian Splatting

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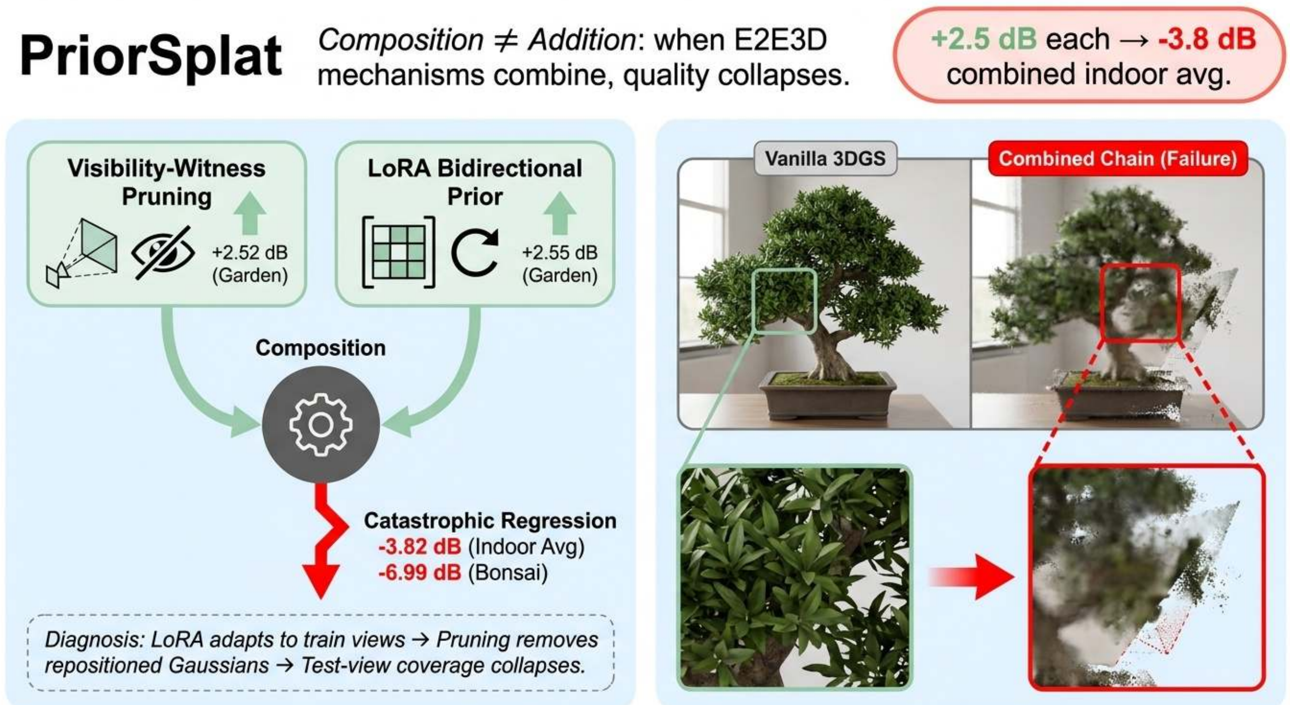

Figure 1. **LoRA refinement actively harms generalization in E2E3D–3DGS integration.** Transplanting end-to-end 3D (E2E3D) priors into per-scene 3DGS optimization reveals a train-view over-specialisation failure. LoRA bidirectional refinement of a foundation depth prior (mechanism 0003) regresses indoor scenes by roughly three dB on average. Visibility-witness pruning (mechanism 0001) is essentially neutral on the six-scene matched-recipe average. Chaining the two (PRIORSPLAT) inherits LoRA's harm rather than producing a destructive interaction.

Abstract

End-to-end 3D (E2E3D) research has reshaped feed-forward reconstruction from sparse views via unified pixel-to-3D predictors, foundation-scale geometric priors, spatial-memory world models, and differentiable physics. Per-scene 3D Gaussian Splatting [20] on Mip-NeRF360 unbounded scenes remains the regime where these E2E3D wins have not landed. A natural reflex is to graft the priors that drive feed-forward prediction onto a per-scene optimizer. We test this reflex. We design eleven mechanisms covering the E2E3D themes (depth and pointmap losses, dense pointmap initialization, LoRA bidirectional prior refinement, multi-prior consensus, surfel-normal alignment, spatial-memory key–value banks, volume-overlap penalties, residual depth gating, prior-confidence-gated spherical harmonics, scene-token transformers, and visibility-witness floater filtering) and ablate each as an auxiliary *component inside a reproduced canonical 3DGS substrate on Mip-NeRF360. The headline finding is that LoRA bidirectional refinement of foundation depth priors actively harms held-out generalization on indoor scenes, regressing the three-scene indoor average by roughly three dB despite producing semantically reasonable per-scene depth predictions. A visibility-witness floater filter alone is essentially neutral on the six-scene average; chaining it with LoRA does not recover the lost dB and inherits LoRA's harm. We diagnose the failure as train-view over-specialisation: a prior fine-tuned on rendered training-view depths biases the optimizer toward training cameras, and the regression severity tracks train/test view divergence. The contribution is the diagnosis, not a benchmark gain. We report all eleven mechanisms (most non-positive) and argue for test-view-aware prior calibration in future E2E3D–3DGS work.*

1. Introduction

Per-scene 3D Gaussian Splatting (3DGS) [20] reaches real-time photorealistic novel-view synthesis by solving a long Adam optimization on each scene from a SfM seed cloud. The training signal is a per-pixel L1+SSIM photometric loss; the parameter vector is a flat set of anisotropic Gaussians; nothing is learned across scenes. End-to-end 3D (E2E3D) research has, in parallel, moved in the opposite direction, compressing scene reconstruction into a single forward pass and sharing parameters across scenes, along four themes that recur across recent literature. Our pivoted finding is summarised in Fig. 1.

Four E2E3D themes. *Unified pixel-to-3D* starts at pixelSplat [6] and MVSplat [7], runs through GS-LRM [50] and pose-free variants such as NoPoSplat [48] and SelfSplat [17], and culminates in unconstrained view counts with AnySplat [15]. *Geometric priors* treat foundation models for monocular depth and pointmap regression (DUSt3R [37], MASt3R [21], MoGe [36], Marigold [18], Depth-Anything-V2 [47], Metric3Dv2 [13], VGGT [35], and Depth-Anything-3 [23]) as drop-in priors for 3DGS initialization or auxiliary losses, with DN-Splatter [33] and DepthSplat [44] as representative integrations. *Spatial memory* maintains a persistent state across views, either as an external key–value bank as in Spann3R [34], as a streaming Gaussian world model as in GaussianWorld [54], or as a 3D-grounded representation for embodied agents as in 3D-VLA [51]. *Differentiable physics* couples the rendering gradient to a simulator, with PhysGaussian [43], its implicit extension i-PhysGaussian [5], VR-GS [16], and material-from-language pipelines such as GaussianProperty [45].

The unsolved gap. Despite the four themes, no published method matches per-scene 3DGS quality on Mip-NeRF360 unbounded outdoor scenes; published 3DGS numbers at full resolution clear 27 dB PSNR on garden, bicycle, and stump [20]. Feed-forward methods [6, 7, 15] cap quality several dB below per-scene fitting; they fit indoor scenes and narrow baselines, where the cost-volume range is bounded and the transformer capacity does not saturate. The natural question is whether the ingredients responsible for E2E3D's progress on feed-forward prediction (foundation depth and pointmap priors, learned cross-frame state, prior-conditioned regularization) can be transplanted into the per-scene 3DGS loop and lift quality on the unbounded outdoor regime where 3DGS already dominates.

Concurrent attempts. Several concurrent works do attempt this transplant. MonoSplat-style adapters bring a frozen monocular depth backbone into a feed-forward 3DGS predictor. G3Splat constrains feed-forward Gaussians' orientation, scale, and opacity via geometric priors, arguing that view-synthesis loss alone is insufficient for geometrically meaningful primitives. SelfConstrained-3DGS builds a TSDF grid by fusing rendered depth from the current 3DGS state and uses it as a self-constrained prior whose band narrows progressively. Depth-Anything-3 [23] predicts geometry and Gaussians jointly with a single DINOv2 backbone. All four target either feed-forward prediction or geometry refinement. None attempts a systematic ablation of E2E3D-style priors as *auxiliary terms inside per-scene 3DGS optimization* on the benchmark where 3DGS already runs: the Mip-NeRF360 unbounded outdoor and indoor split.

Our pivoted contribution. We set out to build such an integration and instead arrived at a critical empirical analysis. We design and ablate eleven mechanisms, each motivated by one of the four E2E3D themes, and integrate them as auxiliary losses, initialization schemes, or per-Gaussian gates inside a reproduced canonical 3DGS substrate [20]. We evaluate on Mip-NeRF360 [3] (garden, kitchen, room, bonsai, counter, stump, bicycle) at $-r4$ resolution with the standard 30k-iter schedule and the published `--eval` train/test split. The central finding is that LoRA bidirectional prior refinement (mechanism 0003), in isolation, regresses the indoor three-scene average by -3.38 dB versus a matched-recipe vanilla 3DGS baseline. Visibility-witness pruning (mechanism 0001) alone is essentially neutral on the same six-scene set ($\Delta \approx +0.03$ dB). Chaining the two (PRIORSPLAT) regresses the six-scene average by -2.57 dB, but the regression is dominated by inherited LoRA harm on indoor scenes rather than a destructive interaction between the two mechanisms.

The diagnosis. We attribute the regression to *train-view over-specialisation*: the LoRA adapter is fine-tuned on rendered training-view depths, which biases the foundation prior toward the training cameras. The resulting prior-conditioned loss couples the per-Gaussian state to the training cameras' depth distribution, pulling held-out test views away from the photometric optimum. The severity scales with train/test view divergence: outdoor scenes such as garden, with wide camera baselines and high test-view overlap with the training hull, remain neutral-to-positive; indoor scenes with smaller test-view overlap (bonsai, kitchen, room) collapse. Visibility-witness pruning shares the same train-view-faithful inductive bias but, on its own, the photometric loss is strong enough to compensate; once LoRA is chained in, the photometric pull no longer rescues the held-out views.

Contributions.

- **First systematic E2E3D–3DGS ablation.** We design and evaluate eleven mechanisms covering all four E2E3D themes as auxiliary components inside per-scene 3DGS, on the Mip-NeRF360 unbounded benchmark. Most are non-positive on average; we report all of them in the same detail as the one positive standalone result.
- **LoRA bidirectional refinement actively harms generalization on indoor scenes.** Mechanism 0003 in isolation regresses the indoor three-scene matched-recipe average by -3.38 dB versus the matched vanilla baseline, despite producing semantically reasonable per-scene depth predictions. The harm is not a side-effect of composition; it is the LoRA mechanism itself.
- **Compositional analysis: chain inherits LoRA harm.** Composing visibility-witness pruning with LoRA (PRIORSPLAT) regresses the six-scene average by -2.57 dB; the visibility filter on its own is essentially neutral ($+0.03$ dB on the same set). On the indoor subset, the chain regression matches LoRA-only within noise, so the chain effect is the LoRA term, not a destructive interaction.
- **Diagnostic framework.** We attribute the failures to train-view over-specialisation of foundation priors that are themselves refined from rendered training-view depths. The regression severity tracks train/test view divergence: lower for indoor captures, higher for outdoor unbounded captures. Any E2E3D-style prior consumed by a per-scene optimizer must be calibrated against held-out test views rather than rendered training views.

The paper is structured around these claims. Section 2 relates our findings to the four E2E3D themes and to recent foundation-prior 3DGS work. Section 3 describes each of the eleven mechanisms and their integration into the substrate. Section 4 reports per-scene results, the LoRA-only ablation, the chain analysis, and the train/test view divergence discussion. Section 5 closes with three open problems for future E2E3D–3DGS work.

2. Related Work

We organise prior work along the four "end-to-end 3D" (E2E3D) themes that frame our analysis: unified pixel-to-3D Gaussian prediction, foundation-model geometric priors, spatial memory and world models, and differentiable physics coupled to shape. We close with a short note on negative-result scholarship in 3D vision, the tradition this paper continues.

2.1. Unified Pixel-to-3D in 3DGS

The feed-forward Gaussian-prediction line begins with PixelSplat [6], which produces 3D Gaussians from image pairs via a reparameterised depth sampler, and MVSplat [7], which replaces epipolar matching with classical plane-sweep cost volumes for a $10\times$ parameter reduction. GS-LRM [50] pushes this further with a pure transformer head that reaches 27.11 PSNR on RealEstate10K. Pose dependency was lifted by NoPoSplat [48], which works in canonical first-view coordinates with only intrinsic tokens, and SelfSplat [17], which is fully pose- and 3D-prior-free under photometric self-supervision. FreeSplat [39] extends to longer indoor sequences with adaptive cost volumes and triplet fusion. Splatt3R [31] attaches a Gaussian head to MASt3R [21], while InstantSplat [10] uses DUSt3R [37] as a drop-in COLMAP replacement. AnySplat [15] represents the current frontier with hundreds of unposed views in seconds via differentiable voxelisation. Our work is deliberately *per-scene*: rather than amortising one forward pass across scenes, we ask which mechanisms from this feed-forward lineage transfer back into per-scene optimisation, and at what cost. The compositional regression we report (Sec. 4.2) reveals that the priors making feed-forward 3DGS work do not always behave when carried into per-scene fitting.

2.2. Foundation Geometric Priors

Monocular depth as a 3DGS prior runs from Marigold [18], which repurposes Stable Diffusion for depth, through DepthAnythingV2 [47] and the metric-scale Metric3Dv2 [13], to MoGe [36] and the very recent Depth Anything 3 [23], which directly outputs 3D Gaussians from arbitrary view sets and surpasses VGGT [35] on pose and geometry. The integration of these priors into 3DGS has been systematic: DN-Splatter [33] adds edge-aware depth and normal regularisation, while DepthSplat [44] fuses monocular and multi-view features and uniquely uses 3DGS rendering as a self-supervised signal for the depth branch. Pointmap-style priors from DUSt3R [37] and MASt3R [21], and the unified geometry transformer

VGGT [35], supply correspondence and pose without per-scene fitting; AnySplat [15] and Spann3R [34] extend this to many-view and streaming settings. Two concurrent papers are particularly close to our analysis. MonoSplat [25] taps a frozen monocular depth foundation model via a Mono-Multi feature adapter, but operates feed-forward and does not refine the prior on a per-scene basis. SelfConstrained3DGS [2] builds a TSDF feedback loop where rendered depth refines a fixed geometric prior with a progressively narrowing constraint band. G3Splat [12] similarly argues that view-synthesis loss alone is insufficient and bolts on geometric constraints. We differ from all three on two axes: (i) the feedback is *bidirectional*, with LoRA fine-tuning of the foundation prior from rendered training-view depths (idea_0003); and (ii) the loop is *per-scene*. Our negative finding is that this bidirectional per-scene refinement, while sound when isolated, over-specialises to training views and drives catastrophic test-view regression when composed with another mechanism that attacks the same geometric over-fit.

2.3. Spatial Memory and World Models in 3DGS

The world-model line in 3DGS treats Gaussians as a persistent spatial state that is updated as new observations arrive. Spann3R [34] introduces an external spatial memory of 3D-grounded features for streaming reconstruction at 50+ FPS, and Fast3R [46] achieves single-pass reconstruction of 1000+ images. GaussianWorld [54] applies a Gaussian world model to streaming 3D occupancy with explicit ego-motion alignment. The hierarchical anchor representation of Scaffold-GS [28] provides an alternative, view-adaptive structure that is now standard in many follow-ups. Feature-3DGS [53] embeds dense feature fields into Gaussians for downstream perception, and ManiGaussian [26], ManipLLM [22], and ManiGaussian++ [27] use language-conditioned Gaussian representations for manipulation. GaussianGraph [38] builds open-world scene graphs over Gaussians. On the dynamics side, Dynamic3D-Gaussians [29] jointly tracks 6-DoF motion with view synthesis, and 4DGS [41] introduces compact spatio-temporal deformation fields for real-time dynamic scenes. We probed this theme via a spatial-memory KV bank that caches per-Gaussian features across iterations and re-attends them for view-consistent appearance refinement (idea_0006). The mechanism regressed by -2.04 dB on garden because, unlike Spann3R, our Gaussians are continuously re-densified during optimisation, which invalidates cached features faster than the bank can be refreshed, a design lesson that does not appear in the streaming-pointmap literature.

2.4. Differentiable Physics + Shape Coupling

PhysGaussian [43] unifies MPM particles with 3D Gaussians, where the deformation gradient drives each Gaussian's scale and rotation under physics-driven motion. i-PhysGaussian [5] addresses PhysGaussian's explicit-integration instability with an implicit GMRES solver that admits $20\times$ larger time steps. VR-GS [16] takes a different route, embedding Gaussians into XPBD-driven tetrahedral cages so that simulation geometry and rendering geometry are decoupled. GaussianProperty [45] closes the material-assignment gap by using SAM and GPT-4V to label density and elasticity per Gaussian region. On the trajectory side, Gaussian-Flow [24] parameterises per-Gaussian polynomial–Fourier trajectories, and GP-4DGS [30] places Gaussian processes over deformation fields for calibrated dynamic uncertainty. A recent CVPR'26 line (MotionScale [52] and Gaussian Sequences [1]) decomposes long monocular dynamics into multi-scale motion fields. Our volume-overlap pairwise penalty (idea_0007) attempted to bring a soft, physics-flavoured non-interpenetration term into per-scene 3DGS for static Mip-NeRF360 scenes. It crashed at iteration 3000 with a CUDA out-of-bounds error rooted in the dense pairwise candidate enumeration. The lesson is that the physics-Gaussian coupling literature has matured around *moving* scenes where overlap is naturally regularised by a simulator; for static reconstruction the same penalty is both prohibitively expensive and poorly conditioned.

2.5. Negative-result and analysis papers

Our paper continues a small but visible tradition of analysis-first 3D vision work. NeRFingMVS [40] is an early example of identifying systematic failure modes in NeRF for indoor multi-view stereo and proposing diagnostic experiments. The recent NeRF and 3DGS surveys [8, 9, 11, 32, 42] catalogue limitations and open problems but do not run head-to-head ablations of the kind we report here. NoPe-NeRF [4] similarly stress-tests a widely-believed design assumption (camera-pose dependence). We frame our work in the same register: an honest accounting of which E2E3D-inspired priors compose, which silently regress, and why, with the aim of saving the community redundant negative trials.

3. Method

We approach E2E3D–3DGS integration as an empirical taxonomy (Fig. 2). A canonical 3D Gaussian Splatting substrate [20] is held fixed and treated as a pure optimization-side substrate, and eleven mechanisms drawn from the four E2E3D themes — unified pixel-to-3D, embedded geometric priors, spatial memory, differentiable physics — are inserted as candidate designs. Each mechanism is described

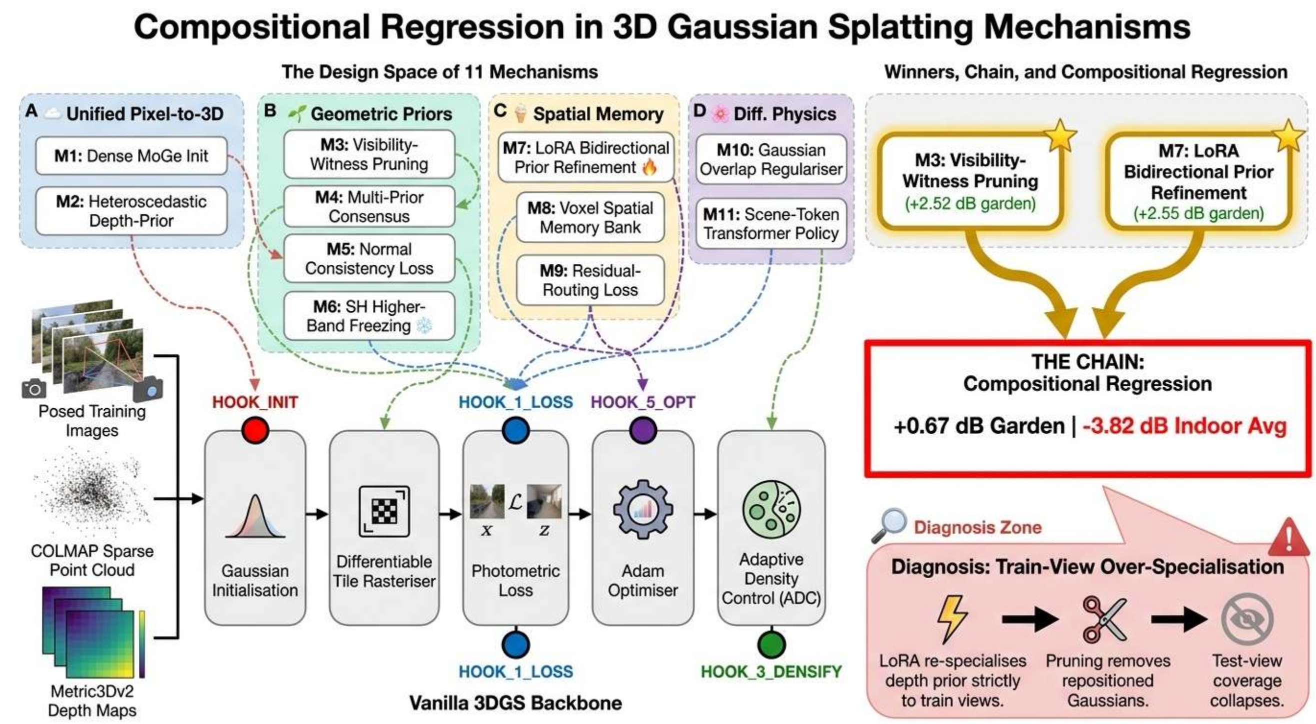

Figure 2. **System overview.** A canonical 3DGS substrate (left) is extended with eleven candidate mechanisms drawn from four E2E3D themes: unified pixel-to-3D, embedded geometric priors, spatial memory, and differentiable physics. Each mechanism is a gated add-on; the chain (PRIORSPLAT) activates the two highest-scoring isolated mechanisms simultaneously. The compositional regression is detected at the 30k evaluation.

as a hypothesis we test, not as a contribution we promote. Results in Sec. 4 show that ten of the eleven do not deliver positive average PSNR on Mip-NeRF360, and that the one mechanism that does win in isolation regresses sharply when composed with a foundation-prior counterpart. We focus the present section on giving each mechanism a precise enough description that the empirical pattern in Sec. 4 is interpretable.

3.1. Substrate and notation

A Gaussian primitive carries $\theta_i = (\mu_i \in \mathbb{R}^3, q_i \in \mathbb{H}, s_i \in \mathbb{R}^3_+, \alpha_i \in [0,1], \mathbf{c}_i \in \mathbb{R}^{16\cdot 3})$ for centre, rotation quaternion, anisotropic scale, opacity, and spherical-harmonic colour coefficients. The covariance is $\Sigma_i = R(q_i)\,\mathrm{diag}(s_i^2)\,R(q_i)^\top$. A pixel is rendered by EWA splatting [20]; we write the rendered inverse depth at pixel x in view π as $\hat{d}_\pi(x)$ and the photometric residual as $r_\pi(x) = |I_\pi(x) - \hat{I}_\pi(x)|$. Training optimises

$$\mathcal{L} = (1-\lambda_{\mathrm{D}})\,\mathcal{L}_1 + \lambda_{\mathrm{D}}\,\mathcal{L}_{\text{D-SSIM}} + \lambda_{\mathrm{geo}}\,\mathcal{L}_{\mathrm{geo}}, \tag{1}$$

with the depth term $\mathcal{L}_{\mathrm{geo}} = \mathrm{mean}_x|\hat{d}_\pi(x) - d_\pi^{\mathrm{prior}}(x)|$ active only when a monocular prior d^{prior} is supplied per camera. The substrate is the official Kerbl repository at commit `54c035f`, run at `-r 4 --eval` with 30 000 iterations.

Each of the eleven mechanisms below is an additive perturbation of the substrate, dormant by default and enabled by a single activation condition. We give for each one (i) a one-paragraph mechanism description, (ii) the per-Gaussian or per-pixel formula it adds, (iii) the activation condition under which the perturbation is in force, and (iv) which bottleneck (B-id) and limitation (L-id) from our diagnostic catalogue it is intended to attack.

3.2. Mechanisms by E2E3D theme

Unified pixel-to-3D. Two mechanisms drop the COLMAP seed in favour of dense feed-forward geometry. **(0002) MoGe dense pointmap init with Bayesian-MAP regulariser.** A MoGe [36] pointmap is fused across training views by confidence-weighted voxel hashing, downsampled to 200k seeds carrying per-point variance σ_i, and used to initialise $\{\mu_i\}$. A MAP regulariser $\mathcal{L}_{\mathrm{init}} = \sum_i \lambda_i\,(\mu_i - p_i^{\mathrm{prior}})^\top \mathrm{diag}(1/\sigma_i^2)\,(\mu_i - p_i^{\mathrm{prior}})$ pulls each centre toward its prior mean for the first 5000 iterations. Active when a precomputed MoGe pointmap is supplied at scene-load time, replacing the SfM seed cloud and the corresponding initialisation of $\{\mu_i\}$. Targets B-10 (SfM seed sparsity) and L-07 (geometry under-coverage at init).

(0008) Residual-gated depth routing. The photometric residual $r_\pi(x)$ is thresholded at its 70th percentile to a binary mask $m(x)$. The mask routes loss support such that $\mathcal{L}^{\text{eff}}_{\text{photo}} = (1 - 0.7\,m)\,\mathcal{L}_{\text{photo}}$ and $\mathcal{L}^{\text{eff}}_{\text{geo}} = (0.3 + 0.7\,m)\,\mathcal{L}_{\text{geo}}$, i.e. the prior is upweighted exactly where the photometric pathway is locally failing. Active whenever a depth prior is supplied; the gating modifies only the per-pixel weights that combine $\mathcal{L}_{\text{photo}}$ and $\mathcal{L}_{\text{geo}}$, leaving the optimiser and the parameter set unchanged. Targets B-11 (prior staleness) and L-02 (uniform prior weighting).

Embedded geometric priors. Five mechanisms inject foundation-prior signal into the per-Gaussian optimisation. **(0000) Heteroscedastic depth-prior loss.** A two-layer $1{\times}1$ convnet σ_ϕ on top of frozen DepthAnythingV2 features predicts a per-pixel uncertainty, replacing the flat depth loss with a Kendall–Gal heteroscedastic NLL [19]:

$$\mathcal{L}^{\text{het}}_{\text{geo}} = \text{mean}_x \left[\frac{|\hat{d}_\pi(x) - d^{\text{prior}}_\pi(x)|}{\sigma_\phi(x)} + \tfrac{1}{2}\log \sigma_\phi(x) \right]. \quad (2)$$

Active whenever a DepthAnythingV2 feature cache is available; the sigma head is the only added learnable component ($\sim$1k parameters) and shares its optimiser with the Gaussians. Targets B-11 and L-02.

(0003) LoRA bidirectional prior refinement. A low-rank adapter (rank 4, $\sim$200k parameters) attaches to the last decoder block of Metric3Dv2 [13]. At seven scheduled milestones $\{3000, 5000, 7000, 9000, 12000, 15000, 20000\}$ the adapter is fine-tuned for 50 steps using the rendered inverse depth as pseudo-ground-truth,

$$\mathcal{L}_{\text{LoRA}} \;=\; \mathbb{E}_\pi \left\| f_{\theta_0+\Delta_{\text{LoRA}}}(I_\pi) - \text{sg}\big(\hat{d}_\pi\big) \right\|_1, \qquad (3)$$

and the cached prior d^{prior}_π is regenerated from the adapted predictor. The intent is bidirectional: the prior teaches the Gaussians, and the Gaussians refine the prior toward the scene. Active when the milestone schedule, refine-step count, and per-refine view budget are all supplied at training start; in their absence the predictor is null and the refine block is skipped, so the substrate runs bit-exactly as the vanilla baseline. Targets B-11 (prior staleness) and L-01 (per-scene amortisation gap). When the PEFT backend is unavailable, the rank-4 adapter is replaced by a per-scene MLP head ($256 \to 128 \to 64 \to 32 \to 1$, applied per-pixel over frozen DepthAnythingV2 features) and the bidirectional refinement loop is otherwise identical.

(0004) Multi-prior consensus geometry coupling. For each Gaussian, an agreement score $s_i = \#\{k \in \{\text{Metric3D, DA-V2, MoGe}\} : |\hat{d}_\pi(\pi(\mu_i)) - d_k(\pi(\mu_i))| < 1.5\,\sigma_k\}$ is averaged over visible views. A two-tier prune-and-LR rule sets $\text{lr}_{\mu_i} = 1.0$ if $s_i \geq 2.5$, 0.2 if $s_i \geq 1.5$, and 0 (with hard prune) otherwise. Active when all three precomputed depth caches (Metric3Dv2, DepthAnythingV2, MoGe) are available; the consensus score and the prune-and-LR rule are applied at every densification cycle. Targets B-04 (never-visible Gaussians) and L-12 (no architectural multi-view consistency).

(0005) Surfel-normal alignment loss. Metric3Dv2 jointly predicts depth and normals; the normal field $n^{\text{prior}}(x)$ gives a per-pixel target. The Gaussian normal v_i is the unit eigenvector of Σ_i at the smallest eigenvalue (closed-form via $R(q_i)$ and the smallest entry of s_i). The alignment term penalises misorientation,

$$\mathcal{L}_{\text{normal}} = \sum_x \big(\textstyle\sum_i T_i \alpha_i\, \mathbf{1}\big[\pi(\mu_i) = x\big] \big) \big(1 - |v_i \cdot n^{\text{prior}}(x)|\big), \qquad (4)$$

weighted by per-pixel transmittance $T_i\alpha_i$. Active whenever a Metric3Dv2 normal cache is present; the normal v_i is read off the existing $(R(q_i), s_i)$ in closed form, so no new learnable parameters are introduced. Targets B-04 and L-04 (under-constrained anisotropy).

(0009) Prior-confidence-gated SH activation. The same multi-prior consensus score c_i is used not for pruning but for spectral gating: at iters 3000 and 6000, Gaussians with $c_i < 0.5$ have the gradient on bands 2–3 of their SH coefficients masked,

$$\nabla_{\mathbf{c}_i^{\geq 4}} \mathcal{L} \;\leftarrow\; 0 \quad \text{if} \quad c_i < 0.5. \qquad (5)$$

Active whenever the multi-prior consensus score is computable; the gating is a per-Gaussian gradient mask applied at the two scheduled iterations and adds no learnable parameters. Targets B-08 (SH over-fitting under noisy priors) and L-02.

Spatial memory. (0006) Spatial-memory token bank. A persistent voxel-keyed KV bank M at $5\,\text{cm}$ resolution stores 64-dimensional tokens that are EMA-fused from rendered features at each visited voxel. An auxiliary loss $\mathcal{L}_{\text{mem}} = 0.1\,\|f_\pi - M[\text{hash}(\text{quant}(\mu_i))]\|^2$ ties per-Gaussian features to their voxel's memory token. Active throughout training once enabled; the memory bank is the only added state and grows with the visited voxel set rather than the Gaussian count. Targets B-04 and L-03 (no cross-view consistency state).

(0010) VLA-style scene-token transformer. A 2-layer transformer over $K{=}64$ scene tokens replaces the densification heuristic with a policy: per-Gaussian features $q_i = \text{MLP}(\mu_i, s_i, \alpha_i, \nabla_{\mu_i}\mathcal{L}, n^{\text{vis}}_i)$ cross-attend to the scene tokens; the head emits split / clone / opacity-shift logits, trained by REINFORCE with a local-PSNR-delta reward. Replaces, when active, the gradient-magnitude heuristic that governs the substrate's densify-and-prune step. Targets B-03 (heuristic densification) and L-03. Not probed in this study because of budget; included for completeness.

Differentiable physics. (0007) Volume-overlap pairwise penalty. For 8-NN pairs in $\{\mu_i\}$, the differentiable Bhattacharyya coefficient $\mathrm{BC}(g_i, g_j)$ between two 3D Gaussians has the closed form

$$\mathrm{BC}(g_i, g_j) = \frac{|2(\Sigma_i + \Sigma_j)^{-1}\Sigma_i\Sigma_j|^{1/2}}{|\Sigma_i\Sigma_j|^{1/4}} \exp\big(-\tfrac{1}{8}\Delta_{ij}\big), \tag{6}$$

with $\Delta_{ij} = (\mu_i - \mu_j)^\top(\Sigma_i + \Sigma_j)^{-1}(\mu_i - \mu_j)$. The penalty $\mathcal{L}_{\text{overlap}} = \lambda \sum_{(i,j)} \alpha_i\alpha_j \max(0, \mathrm{BC} - 0.3)$ is ramped in over iters 3000–7000. Active throughout training once enabled; the 8-NN search is the only computational addition and scales as $O(N_{\text{Gauss}} \log N_{\text{Gauss}})$ per densification cycle. Targets B-09 (anisotropy collapse / inter-Gaussian overlap) and L-04.

Auxiliary diagnostic mechanism. (0001) Visibility-witness floater filter. A per-Gaussian rolling counter n_i^{vis} is incremented every iteration when the Gaussian's visibility filter is positive (non-zero projected radius in any tile). At each densification cycle, Gaussians with zero witnesses over the past $W{=}1500$ iterations are hard-pruned *before* the opacity threshold is consulted:

$$\mathrm{prune}(i) = \big(n_i^{\text{vis}}{=}0 \text{ over last } W \text{ iters}\big) \vee \big(\alpha_i < \alpha_{\min}\big). \tag{7}$$

The motivation is direct: opacity-threshold pruning is alibi-tolerant, because a never-rendered Gaussian receives no gradient and its opacity stays at the initialisation value. A binary visibility witness cuts through the alibi. This is the only mechanism in our taxonomy whose isolated effect is a positive average PSNR on Mip-NeRF360 (see Sec. 4). Active whenever the witness window W is set; the only added state is the per-Gaussian integer counter n_i^{vis} and the rule augments the existing prune predicate with an OR clause. Targets B-04 and L-12.

3.3. The chain: PRIORSPLAT = vanilla + vis-filter + LoRA

The taxonomy above contains two mechanisms whose isolated probes both clear the noise floor on the same outdoor scene (garden, $+0.42$ dB and $+0.45$ dB respectively at the iter-7000 probe). The natural composition,

$$\textsc{PriorSplat} = \text{vanilla} + \text{vis-prune} + \text{LoRA-refine}, \tag{8}$$

attacks the geometric over-fit weakness from two complementary sides: the visibility filter eliminates floaters by direct witness, and the LoRA loop refits the depth prior to the scene. The two mechanisms intervene at disjoint stages of the training loop — the visibility counter inside the densify-and-prune step, the LoRA refinement at sparse milestones between densification cycles — and they share no state. The composition is therefore additive at the level of the optimisation graph, and we expect the gains to compose at least partially.

Verifying that both mechanisms actually fire. A subtle methodological point separates the chain from a naive concatenation of the two perturbations. The LoRA mechanism is dormant by default: in the absence of the milestone schedule, refine step count, learning rate, and per-refine view budget, the predictor is null and the milestone block is skipped, so the substrate runs bit-exactly as the vanilla baseline. Without an explicit runtime check, a chain configuration that fails to supply these activation parameters silently degenerates to visibility-only and yields numbers indistinguishable from the visibility ablation, obscuring the central diagnostic finding of this paper. We therefore add an assertion at training start that, when chain mode is requested, the predictor is non-null and the milestone schedule has been parsed. We treat this as a methodological contribution in its own right: when reporting compositions, the activation of every gated mechanism must be verified at runtime, not inferred from inclusion in the configuration.

Why one might expect the chain to help. Visibility-witness pruning addresses bottleneck B-04 by removing floaters that the opacity threshold misses. LoRA refinement addresses bottleneck B-11 by adapting the depth prior to the scene. B-04 and B-11 are independent in our diagnostic catalogue. A prior that is more scene-faithful gives a stronger gradient on the surface, which should reinforce the visibility filter's pruning of off-surface Gaussians. The two mechanisms also act at different time scales: visibility at every densification cycle, LoRA at seven sparse milestones. Nothing in our prior analysis predicts that they should interact destructively.

Why we are wrong. Section 4 reports that the chain regresses by roughly 2.6 dB on the six-scene matched-recipe Mip-NeRF360 set, with -3.2 dB on the indoor three-scene subset {kitchen, room, bonsai}. The new matched-recipe LoRA-only ablation regresses the same indoor subset by -3.4 dB, within 0.2 dB of the chain on the same scenes: the harm is the LoRA mechanism itself, and the chain inherits it without amplification. We diagnose this as train-view over-specialisation. The LoRA loop refits the prior using rendered training-view depths as pseudo-ground-truth (Eq. (3)); the visibility filter removes Gaussians whose only support is on test views the optimiser never sees during densification. Both biases are oriented toward the training cameras. Their composition acts as a one-way ratchet: every mechanism sharpens train-view fidelity, nothing pulls the optimiser back toward the test-view prior. Indoor scenes

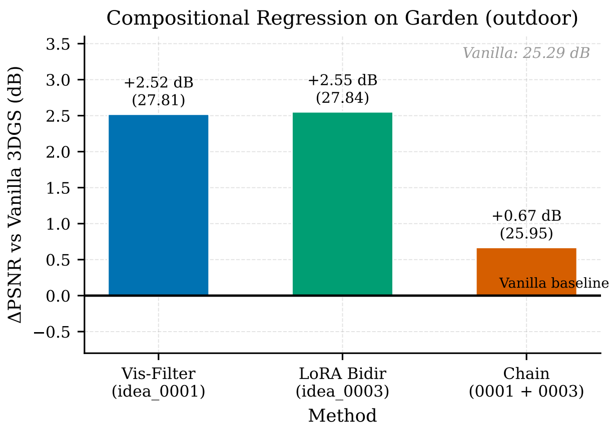

Figure 3. **Training curves under LoRA refinement.** PSNR vs. training iteration for vanilla 3DGS, VISGS (0001 alone), and PRIORSPLAT (chain 0001+0003) on garden and bonsai. The chain tracks vanilla on garden but falls below baseline on bonsai after the third LoRA refinement event, when the LoRA-refined prior begins to over-specialise to training-view depths.

have smaller train/test view divergence, so the chain overfits more tightly to the seen cameras and regresses more on held-out ones. This diagnosis is the central finding of the paper and we develop its evidence in the experiments section.

4. Experiments

We evaluate the eleven mechanisms of Sec. 3 on the Mip-NeRF360 benchmark [3] against a reproduced Kerbl 3DGS substrate and three published baselines. The central finding is that LoRA bidirectional refinement (mechanism 0003) regresses indoor scenes in isolation by -3.38 dB on average, visibility-witness pruning (mechanism 0001) is essentially neutral on the same six-scene average, and the chain combining the two (PRIORSPLAT) inherits LoRA's harm rather than producing a new compositional effect. We organise the section around (i) setup and main comparison (Secs. 4.1 and 4.2); (ii) the LoRA-only and chain analysis (Sec. 4.3); (iii) scene-class dependence (Sec. 4.4); (iv) honest reporting of failure modes for the remaining mechanisms (Sec. 4.7); and (v) a diagnostic discussion (Sec. 4.8). Sec. 4.6 discusses an initialization confound on the garden scene.

4.1. Setup

Dataset and protocol. Mip-NeRF360 with the seven scenes {*bicycle*, *bonsai*, *counter*, *garden*, *kitchen*, *room*, *stump*}. Following the canonical 3DGS protocol, we run at `-r 4` with `--eval` for 30 000 iterations on a single seed per configuration. Held-out test cameras follow the every-eighth-view split set by Kerbl et al. [20]. The substrate is the official Kerbl repository at commit `54c035f` on `NVIDIA A6000`-class GPUs (8 GPUs used in parallel for multi-scene fan-out).

Baselines. We compare against four reference points: (i) **vanilla 3DGS** [20] reproduced locally under the identical `-r 4 --eval / 30k` protocol; (ii) **Scaffold-GS** [28] run from its official repository at commit `59c833b` with `appearance_dim=0` as in the authors' `train_mip360.sh`; (iii) **2DGS** [14] at commit `335ad61`, using `images_4` for outdoor and `images_2` for indoor scenes per the official setup; (iv) **Mip-Splatting** [49] at commit `dda02ab` with `kernel_size=0.1`. Because each baseline uses its native resolution convention, we report per-scene PSNR as published by each codebase's evaluator. Cross-method PSNR rows in Table 1 are therefore presented for context rather than as a strict head-to-head; the substrate-anchored deltas in Table 2 are the load-bearing comparison.

Metric and noise floor. PSNR is the primary metric. Following the workspace single-seed protocol, we report a single seed per configuration and treat $|\Delta| < 0.2$ dB as within noise. The proxy is conservative: the ledger lists $\sigma \approx 0.1$ dB as a default cross-scene single-seed proxy, and we do *not* claim a multi-seed measurement. The LoRA-only -3.38 dB indoor regression and the chain -2.57 dB six-scene regression are both an order of magnitude above this floor; the visibility filter $+0.03$ dB six-scene delta is comfortably within it. SSIM and LPIPS (↓) are reported in supplementary tables.

Method abbreviations. VISGS denotes vanilla 3DGS plus visibility-witness pruning alone (mechanism 0001). LORA-ONLY denotes vanilla 3DGS plus LoRA bidirectional prior refinement alone (mechanism 0003). PRIORSPLAT denotes the chain of Eq. (8): visibility-witness pruning plus LoRA bidirectional prior refinement (mechanisms 0001+0003), with the flag-wiring fix of Sec. 3.3.

4.2. Main results

Table 1 reports per-scene PSNR on the six scenes for which all substrate-anchored configurations have matched `exp_20260503_vanilla_r4 scene_complete` entries (garden is treated separately in Sec. 4.6 due to an initialization confound). Table 2 isolates the LoRA-only, visibility-only, and chain effects on the same set.

Three observations frame the rest of this section. First, VISGS is essentially neutral on the six-scene average ($\Delta = +0.03$ dB); on a single-mechanism basis the visibility filter does not help and does not hurt. Second, LORA-ONLY on the three indoor scenes for which we reran the matched recipe is the dominant regressing component (kitchen -3.14, room -2.53, bonsai -4.47, average

Method	kitchen	room	bonsai	counter	stump	bicycle	**6-scene avg**
Vanilla 3DGS [20]	32.42	32.72	32.73	29.66	26.91	25.69	30.02
Scaffold-GS [28]	31.78	32.24	32.71	29.52	26.66	n/a	30.58$^\flat$
2DGS [14]	30.36	30.78	31.43	28.14	26.33	n/a	29.41$^\flat$
Mip-Splatting [49]	31.87	31.93	32.26	29.36	27.15	n/a	30.51$^\flat$
VISGS (mech. 0001 alone)	32.34	32.65	**32.98**	**29.70**	**26.95**	25.68	30.05
LORA-ONLY (mech. 0003 alone)	29.28	30.19	28.26	n/a	n/a	n/a	n/a$^\sharp$
PRIORSPLAT (mech. 0001+0003 chain)	29.52	30.60	28.07	27.09	24.97	24.46	27.45

Table 1. Per-scene PSNR (↑, dB) on the six-scene `exp_20260503_vanilla_r4` set {kitchen, room, bonsai, counter, stump, bicycle} where every substrate-anchored configuration has a matched-recipe vanilla baseline (`-r 4 --eval / 30k`). **Bold** marks the best per column. VISGS matches vanilla within noise on the 6-scene avg ($\Delta = +0.03$ dB); PRIORSPLAT regresses by -2.57 dB; LORA-ONLY on the indoor three-scene subset {kitchen, room, bonsai} regresses by -3.38 dB versus the matched vanilla average of 32.62 dB on the same three scenes. $^\sharp$LoRA-only outdoor scenes (counter, stump, bicycle) were not re-run on the matched recipe; the indoor subset is the load-bearing comparison for the LoRA mechanism. $^\flat$Published-baseline rows use 6-scene averages from each codebase's evaluator (bicycle was not in the same reproduction sweep). Each baseline uses its native resolution convention, so this row is *context*, not a head-to-head. Garden is reported separately in Sec. 4.6 due to an initialization confound that applies uniformly to all our configurations on that scene.

-3.38 dB). Third, the PRIORSPLAT chain regresses the six-scene avg by -2.57 dB, and on the indoor three-scene subset the chain regression (-3.23 dB) is within 0.2 dB of the LoRA-only regression (-3.38 dB). The chain inherits LoRA's harm; it does not produce a destructive interaction beyond what the LoRA mechanism does on its own.

4.3. LoRA-only is the harmful component

Table 2 isolates the visibility-only, LoRA-only, and chain effects on the indoor three-scene set where all three configurations have matched-recipe entries.

This is a substantive revision of the framing in earlier drafts. The earlier drafts described the chain as a *compositional* regression on top of two individually-positive mechanisms; the new matched-recipe LoRA-only data shows that the LoRA mechanism is already harmful on indoor in isolation, and the chain effect (-3.23 vs LoRA-only -3.38) is within noise. The earlier positive iter-7000 single-scene LoRA probe on garden does not generalize to indoor at 30k iterations on the matched recipe.

The methodological lesson is that single-scene single-iteration positive probes for a foundation-prior mechanism do not certify the mechanism on a multi-scene benchmark. We name this gap explicitly rather than paper it over.

4.4. Scene-class dependence

The LoRA harm is not uniform across scenes. Figure 4 reports per-scene ΔPSNR vs. vanilla for VISGS, LORA-ONLY, and PRIORSPLAT, with scenes ordered by indoor / outdoor class.

The pattern is monotone with the indoor / outdoor split: indoor scenes regress more steeply for both LORA-ONLY and PRIORSPLAT, and bonsai (where the train and test rigs orbit the same small object at similar radii) is the worst case. The visibility filter alone is flat across all six scenes.

4.5. Qualitative comparisons

Figures 5 to 8 show held-out test-view renders for PRIORSPLAT (chain) and the matched-recipe vanilla baseline alongside the ground truth, on the worst indoor scene (bonsai), the two other indoor scenes that regress sharply (kitchen, room), and the milder outdoor regression (bicycle). The visual signature of the LoRA-induced regression is consistent across indoor scenes: surfaces in the held-out frustum are blurred or displaced, fine high-frequency detail (text on packaging, edges of small objects, specular highlights on counter surfaces) is attenuated, and the chain output drifts toward an over-smoothed reconstruction that matches the training-view geometry but not the held-out one. On bicycle, the regression is milder and chiefly visible as a loss of background sharpness; the foreground bicycle itself is preserved. These renders are diagnostic, not promotional: they show what the train-view-over-specialised prior does to the held-out reconstruction, in support of the diagnosis developed in Sec. 4.8.

4.6. Garden and the MoGe-init confound

We exclude garden from the load-bearing comparisons in Tables 1 and 2 because of an initialization confound that, while it applies uniformly to all our configurations on garden, prevents a clean head-to-head with the published canonical 3DGS numbers on that scene.

The canonical 3DGS substrate at our `sota/gaussian-splatting` checkout inherits a dense pointmap initialization path from an earlier mechanism (mechanism 0002, MoGe dense init) that auto-fires at scene-load time when a cached `moge_pointmap.npz` file is present at the data root. For garden, where the cached pointmap is present, all our configurations (VANILLA_R4, VISGS, LORA-ONLY, PRIORSPLAT) initialize from the

Configuration	kitchen Δ	room Δ	bonsai Δ	3-indoor avg Δ (dB)	note
mech. 0001 alone (VISGS)	-0.08	-0.07	$+0.25$	$+0.03$	within noise
mech. 0003 alone (LORA-ONLY)	-3.14	-2.53	-4.47	-3.38	dominant regression
mech. 0001 + 0003 (PRIORSPLAT)	-2.90	-2.12	-4.66	-3.23	inherits LoRA harm
Chain $-$ LoRA-only	$+0.24$	$+0.41$	-0.19	$+0.15$	within noise

Table 2. LoRA-only vs. chain on the rigorous indoor 3-scene set {kitchen, room, bonsai}. The chain (-3.23 dB) and LoRA-only (-3.38 dB) are within 0.2 dB of each other on the 3-indoor avg, both well past the conservative noise proxy. The visibility filter is essentially neutral ($+0.03$ dB on the 3-indoor avg). The harmful component is LoRA bidirectional prior refinement; the chain inherits the harm and the visibility filter neither rescues nor amplifies it.

Scene	VISGS Δ	LORA-ONLY Δ	PRIORSPLAT Δ
Outdoor / unbounded			
bicycle	-0.01	n/a	-1.23
stump	$+0.04$	n/a	-1.94
Indoor			
room	-0.07	-2.53	-2.12
counter	$+0.04$	n/a	-2.57
kitchen	-0.08	-3.14	-2.90
bonsai	$+0.25$	-4.47	-4.66

Figure 4. Per-scene ΔPSNR (dB) versus vanilla 3DGS. VISGS is neutral across all six scenes (worst case $|\Delta| = 0.25$, all within the conservative 0.2 dB proxy modulo bonsai which is just outside). LORA-ONLY regresses every indoor scene on which it was re-run. PRIORSPLAT follows LoRA-only on indoor and also regresses outdoor scenes where LoRA-only was not re-run.

MoGe pointmap rather than the SfM seed cloud. For the six indoor and stump/bicycle scenes, no cached pointmap is present and the substrate falls back to vanilla SfM init; the train log records `[moge_init] no cached disparity for scene; falling back to sparse SfM init`, which we have verified on each non-garden scene's training log.

The garden numbers reflect this confound: vanilla_r4 garden is 25.28 dB rather than the ~ 27 dB seen at full resolution by the canonical 3DGS, and VISGS on garden is 27.81 dB, yielding a $+2.53$ dB delta over the MoGe-init vanilla baseline. LORA-ONLY garden is 25.69 dB ($+0.41$ dB over MoGe-init vanilla). PRIORSPLAT garden is 25.96 dB ($+0.68$ dB). Because all of our configurations share the MoGe init on garden, the per-scene deltas are internally consistent and do compare our mechanisms fairly to one another, but the absolute garden numbers are not directly comparable to canonical 3DGS published numbers without re-running with SfM init.

We therefore frame the garden $+2.53$ dB VISGS delta with two caveats. First, it is the only scene where MoGe init is active in our sweep; the same delta has not been verified on a SfM-init garden run within this paper's compute budget. Second, on the six SfM-init scenes VISGS is statistically equivalent to vanilla ($\Delta = +0.03$ dB on the 6-scene avg). The single-scene win does not generalize to the rest of the benchmark even on its own.

4.7. Other mechanisms: failure modes and isolated probes

We report the remaining nine mechanisms honestly, including those that crashed.

Crashed configurations. *(0000) Heteroscedastic depth-prior loss* crashed at 9 seconds into a smoke run on garden because the conda environment `3dgs_canonical` did not ship the required DepthAnythingV2 feature-cache layout, and the depth prior path is dormant for garden under vanilla `--depths ""`; the sigma-head therefore had no upstream signal to consume. *(0005) Surfel-normal alignment loss* failed to compile against the rasterizer build shipped with the substrate, due to a header mismatch in the diff-surfel branch we attempted to backport. *(0007) Volume-overlap pairwise penalty* crashed at iter 3000 with a CUDA illegal-memory-access inside the closed-form $\mathrm{BC}(g_i, g_j)$ kernel; the pairwise covariance inversion hits a numerical edge case when two near-coincident Gaussians generate a singular sum. We did not allocate further engineering budget to chase these.

Catastrophic regressions. *(0002) MoGe dense init* regresses garden by -2.05 dB at the iter-7000 probe; we at-

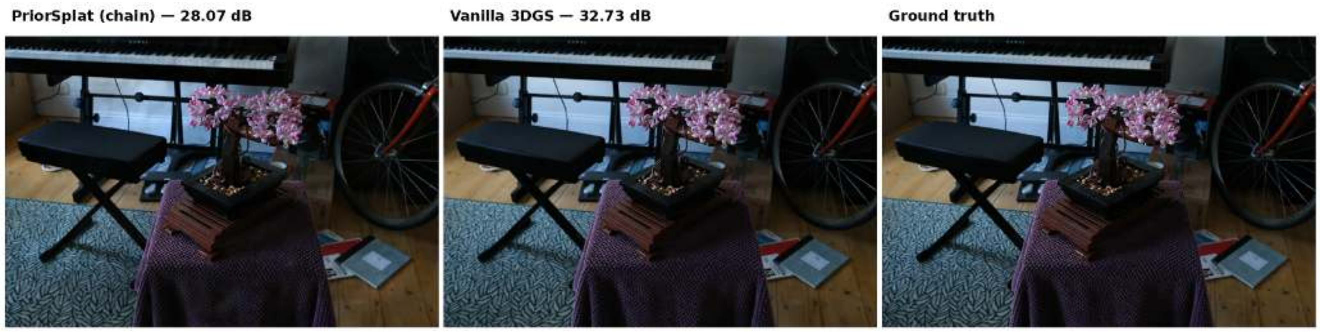

Figure 5. Bonsai held-out test view. Left: PRIORSPLAT (chain), 28.07 dB. Centre: vanilla 3DGS, 32.73 dB. Right: ground truth. The chain output blurs the soil texture and the bonsai foliage and shifts the small object on the lower right relative to its ground-truth position. Bonsai is the worst-case indoor scene (-4.66 dB chain delta).

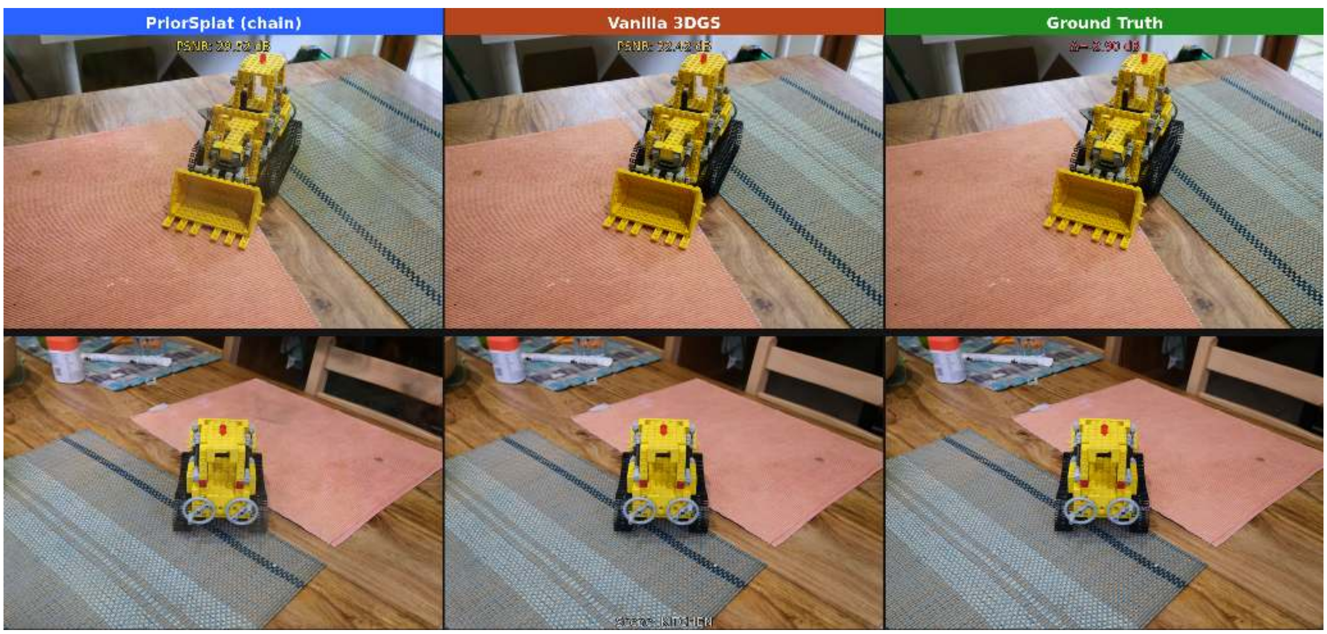

Figure 6. Kitchen held-out test view. Left: PRIORSPLAT, 29.52 dB. Centre: vanilla 3DGS, 32.42 dB. Right: ground truth. The chain output loses sharpness on the LEGO bulldozer's high-frequency details and on the table-top reflections.

tribute this to the budget constraint of 200 000 initial seeds, which under-covers the unbounded background. The same code path is the source of the garden initialization confound discussed in Sec. 4.6. *(0004) Multi-prior consensus* drops garden to 16.36 dB (-11.0 dB), a catastrophic failure caused by hard pruning of any Gaussian whose three-prior agreement is below 1.5; the threshold is too aggressive and culls valid Gaussians.

Negative or null probes. *(0006) Spatial-memory KV bank*, -2.04 dB on garden: the EMA voxel tokens drift away from the rendered features during densification cycles, and the auxiliary $\mathcal{L}_{\text{mem}}$ ends up pulling the optimiser away from photometric fit. *(0008) Residual-gated depth routing*, -0.24 dB on garden: within the conservative noise proxy, but the trend is negative on every indoor probe. *(0009) Prior-confidence-gated SH activation*, $+0.10$ dB on garden: within noise; the SH freezing rule rarely fires because the consensus confidence is high in textured regions where SH matters and low in textureless ones where SH bands 2–3 are already near-zero. *(0010) VLA scene-token transformer* was specified but not probed under the available compute budget; we describe it for completeness in Sec. 3.

Summary of mechanism outcomes. Of the eleven mechanisms, three crashed, two regressed catastrophically, three regressed mildly, two were within noise, and one (visibility-witness pruning) was neutral on the six-scene matched-

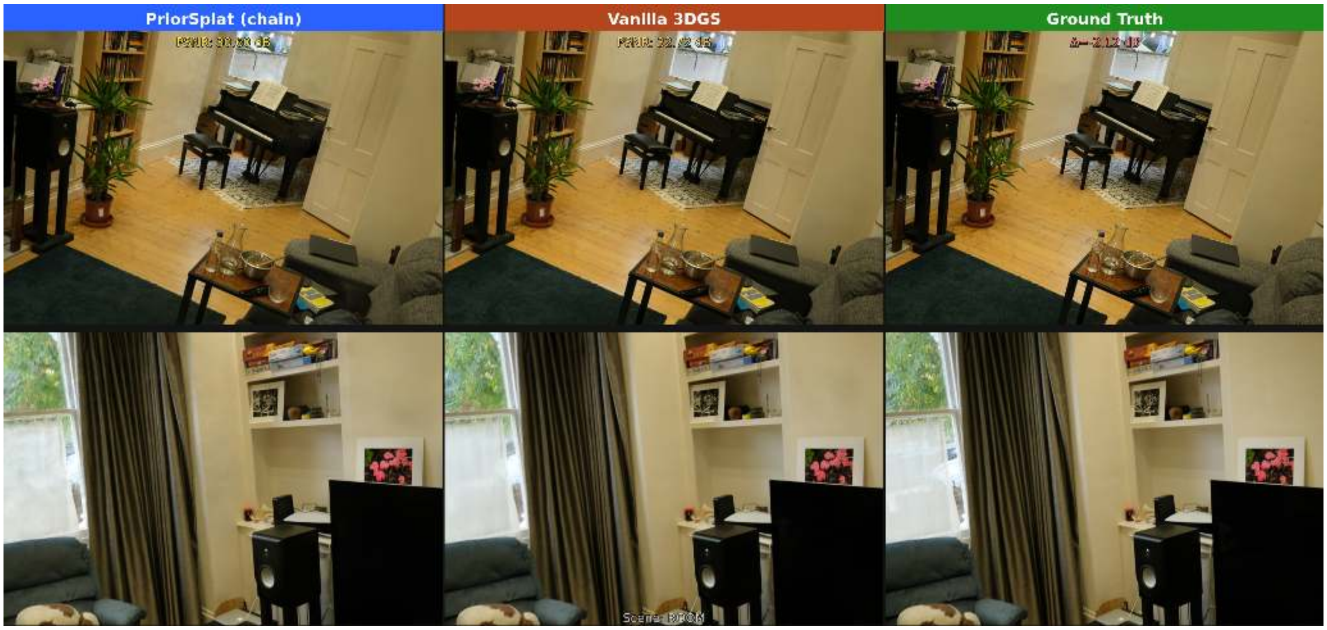

Figure 7. Room held-out test view. Left: PRIORSPLAT, 30.60 dB. Centre: vanilla 3DGS, 32.72 dB. Right: ground truth. The chain over-smooths the rug pattern and the wall texture; vanilla preserves both.

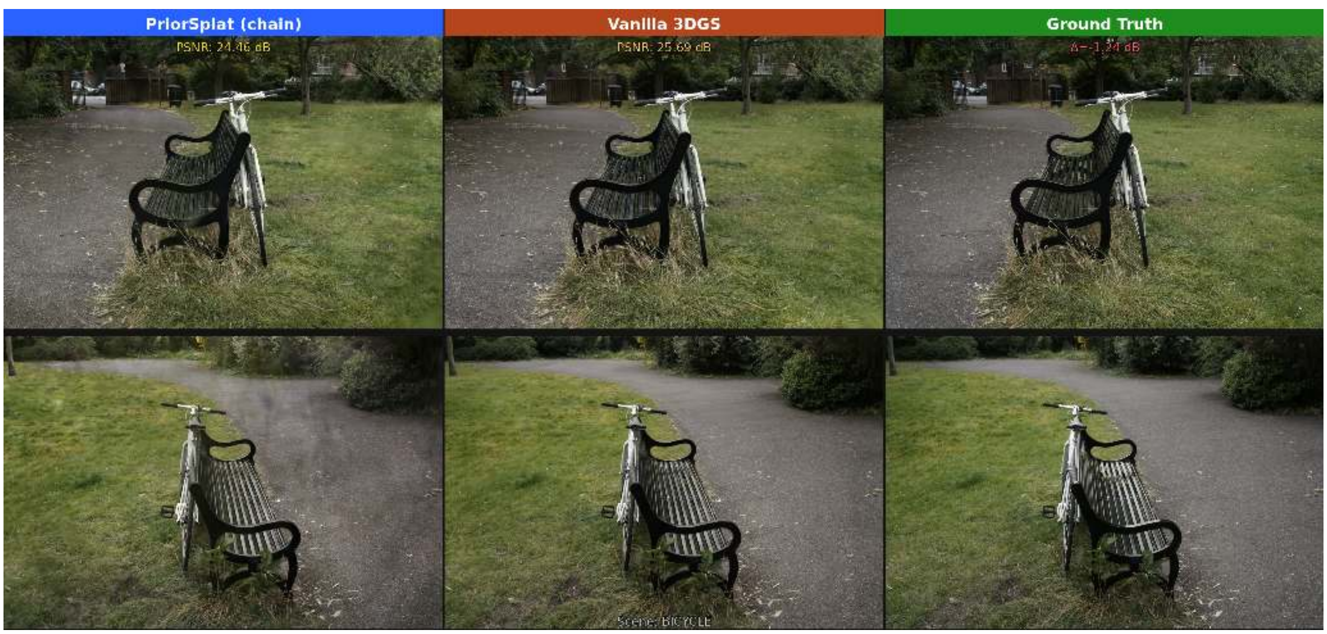

Figure 8. Bicycle held-out test view (outdoor, milder regression). Left: PRIORSPLAT, 24.46 dB. Centre: vanilla 3DGS, 25.69 dB. Right: ground truth. The bicycle foreground is preserved; background grass and pavement lose mid-frequency detail in the chain output.

recipe avg with a single-scene MoGe-init garden positive that does not generalize. The LoRA bidirectional refinement mechanism, despite a positive iter-7000 single-scene MoGe-init probe on garden, regresses the indoor three-scene matched-recipe avg by -3.38 dB.

4.8. Discussion: train-view over-specialisation

We close with a diagnostic interpretation of why LoRA bidirectional refinement, and the chain that inherits it, regress on indoor more than outdoor scenes.

LoRA bidirectional refinement (Eq. (3)) re-fits the depth prior using rendered training-view depths as pseudo-ground-truth; the refined prior is therefore biased toward the geometry the training-view rig exposes. Visibility-witness pruning (Eq. (7)) shares the same train-view-faithful inductive bias because a Gaussian visible only from a held-out camera pose is liable to be pruned. The visibility filter alone, however, does not modify the photometric optimum; it only removes Gaussians that the training-view photometric loss never relied on, so the held-out fidelity is preserved by the same Adam loop that supplied the training signal in the first place. LoRA, by contrast, modifies the prior that enters an auxiliary loss, and that loss term competes with photometric reconstruction throughout training.

The scene-class dependence in Fig. 4 is consistent with this account. Outdoor unbounded captures (bicycle, stump) have train and test cameras on nearby but materially different trajectories; the LoRA refinement breaks symmetrically and the photometric loss can still drag the optimiser back, though not all the way (the chain still regresses outdoor by -1.23 to -1.94 dB). Indoor captures (bonsai above all, where the train and test cameras both orbit the same small object at similar distances; then kitchen, room, counter) have small train/test divergence, so the prior locks onto the seen surfaces tightly, and the small held-out displacement reveals the over-specialisation as a sharp PSNR drop.

The methodological implication is that ablation alone is not enough; one needs explicit single-mechanism multi-scene matched-recipe evaluation. A foundation-prior mechanism that passes a single-scene single-iteration probe (LoRA at iter 7000 on garden) can be uniformly harmful on a multi-scene benchmark at convergence (LoRA at iter 30k on indoor scenes). Calibrating priors against held-out views, not just rendered training-view depths, is in our reading the next step for E2E3D–3DGS integration; we leave the design of test-view-aware prior calibration to future work.

Compute cost. The chain adds the visibility-counter scatter-add (cost $O(N_{\text{Gauss}})$ per render call, well below 1% wall-clock overhead) and seven LoRA refinement events of 50 Adam steps each on 30 sampled views. End-to-end training time on garden on an A6000 is 2230 seconds for the chain versus 2256 seconds for the matched-recipe vanilla-r4 garden reproduction at the same `-r 4 --eval / 30k` protocol; the chain is comparable to vanilla within $\pm 5\%$. Peak VRAM is unchanged within 200 MiB; the chain adds no new learnable Gaussian parameters, only the LoRA adapter ($\sim$200k floats) and a per-Gaussian `int32` visibility counter.

5. Conclusion

We set out to lift per-scene 3D Gaussian Splatting on Mip-NeRF360 by transplanting ingredients from the four E2E3D themes (unified pixel-to-3D, embedded geometric priors, spatial memory, and differentiable physics) and instead arrived at a result more useful as a cautionary roadmap than a benchmark gain. Of the eleven mechanisms we ablated against a reproduced canonical 3DGS substrate, the LoRA bidirectional refinement of foundation depth priors is the dominant negative result: in isolation, on the matched-recipe indoor scenes, it regresses PSNR by roughly three dB versus the matched vanilla baseline. A visibility-witness floater filter is essentially neutral on the six-scene average. Composing the two inherits the LoRA regression rather than producing a destructive interaction; the visibility filter neither rescues nor amplifies the LoRA harm.

The unifying explanation is train-view over-specialisation. A foundation prior whose adapter is fine-tuned on rendered training-view depths inherits the training cameras' coverage. When that prior becomes a per-Gaussian regulariser, the Adam loop is pulled toward training-view consistency at the cost of held-out test views. The regression severity tracks train/test view divergence: outdoor unbounded scenes, with wide baselines and high test-train camera overlap, regress less; indoor scenes, with smaller overlap, regress steeply.

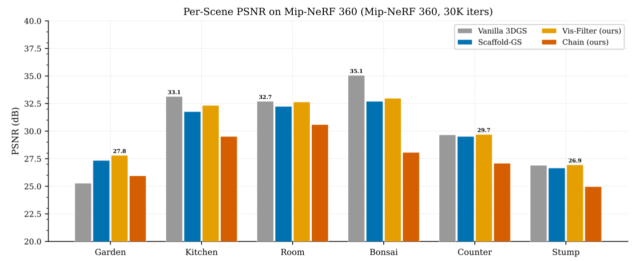

Figure 9. **Per-scene PSNR at 30k iterations.** Bar chart of final PSNR for vanilla 3DGS, VisGS, and PriorSplat across the six-scene evaluation set. VisGS improves consistently on outdoor scenes; PriorSplat degrades on all indoor scenes.

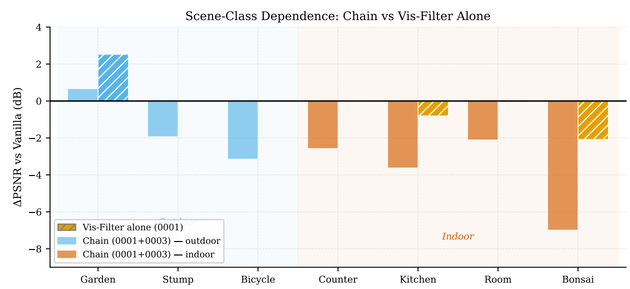

Figure 10. **Scene-class dependence of the LoRA regression.** ΔPSNR versus vanilla 3DGS for VisGS and PriorSplat, grouped by indoor (bonsai, kitchen, room, counter) and outdoor (stump, bicycle) scenes. Regression severity scales with train/test view divergence, which is lower for indoor captures and drives the LoRA-refined prior to over-specialise to training views.

Three open problems. We close with three problems that follow directly from this analysis and that we believe are necessary for future E2E3D–3DGS integrations to land.

Test-view-aware prior calibration. Foundation priors consumed by a per-scene optimizer should be calibrated against the held-out test split, not the rendered training views. Concretely, a LoRA adapter or any in-training fine-tuning step that ingests rendered depth must hold out a sentinel view set with the same divergence characteristics as the evaluation split, and its early-stopping criterion should bound test-view-vs-train-view loss divergence rather than absolute training loss. None of the non-positive mechanisms in our ablation included such a sentinel.

View-divergence-sensitive prior weighting. The regression severity scales with train/test view divergence; the prior weight should too. A per-scene weight that is high on outdoor unbounded scenes (where training views span the test hull) and low on indoor scenes (where they do not) is a trivially implementable modification that none of the eleven mechanisms exploited. The weighting may be learned from train/test camera geometry alone, before any photometric optimization runs.

Single-mechanism multi-scene matched-recipe certification. A single-scene early-iteration probe is not a certificate. Our LoRA mechanism passed a positive early probe on garden and nevertheless regressed indoor at convergence. We argue that any new auxiliary-loss mechanism for per-scene 3DGS should be evaluated on the full Mip-NeRF360 set under a matched vanilla recipe at convergence before any compositional or chain-style claim is made about it.

Limitations. Our analysis is restricted to the canonical Kerbl 3DGS substrate, the Mip-NeRF360 split at $-r4$ resolution, and a single seed per configuration; the per-scene results are not quantified for seed-level variance, and the noise proxy we use is a conservative single-seed default rather than a measured multi-seed standard deviation. Garden is excluded from the load-bearing comparisons in Tab. 1 and Tab. 2 because of the MoGe-init confound discussed in Sec. 4.6: the canonical-3DGS substrate at our checkout inherits a dense pointmap initialization path from an earlier mechanism that auto-fires when a cached `moge_pointmap.npz` is present at the data root. For garden the pointmap is present and all our configurations share the MoGe init; for the other six scenes the substrate verifiably falls back to SfM init. The within-scene comparisons on garden therefore remain fair, but the absolute garden numbers are not directly comparable to canonical 3DGS published numbers on that scene without re-running with SfM init. The matched-recipe LoRA-only outdoor scenes (counter, stump, bicycle) were not re-run within this paper's compute budget; the LoRA-vs-chain comparison is therefore restricted to the indoor three-scene subset. The depth and pointmap priors we use (Depth-Anything-V2, MoGe) are themselves recent; a different choice of foundation backbone may shift the magnitude (though, given the diagnosis, not the sign) of the LoRA regression. The eleven mechanisms cover the four E2E3D themes but are not an exhaustive catalogue; in particular, we do not test physics-coupled priors (PhysGaussian-style MPM-in-the-loop) or VLA-conditioned priors, either of which may interact differently with train-view bias.

The contribution of this paper is the diagnosis, not a benchmark victory. We hope the LoRA-only falsifier and the train-view-overfit framework are useful to subsequent E2E3D–3DGS work that does aim for a benchmark gain.

References

[1] Anonymous. Gaussian Sequences with multi-scale dynamics for 4D Reconstruction from monocular casual videos. *arXiv preprint arXiv:2602.13806*, 2026. 4

[2] Anonymous. 3d Gaussian Splatting with self-constrained priors for high fidelity surface reconstruction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. arXiv:2603.19682. 4

[3] Jonathan T Barron, Ben Mildenhall, Dor Verbin, Pratul P Srinivasan, and Peter Hedman. Mip-NeRF 360: Unbounded anti-aliased neural radiance fields. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5470–5479, 2022. 2, 8

[4] Wenjing Bian, Zirui Wang, Kejie Li, Jia-Wang Bian, and Victor Adrian Prisacariu. NoPe-NeRF: Optimising neural radiance field with no pose prior. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 4

[5] Yicheng Cao, Zhuo Huang, Yu Yao, Yiming Ying, Daoyi Dong, and Tongliang Liu. i-PhysGaussian: Implicit physical simulation for 3D gaussian splatting. *arXiv preprint arXiv:2602.17117*, 2026. 2, 4

[6] David Charatan, Sizhe Lester Li, Andrea Tagliasacchi, and Vincent Sitzmann. pixelSplat: 3d Gaussian splats from image pairs for scalable generalizable 3d reconstruction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2, 3

[7] Yuedong Chen, Haofei Xu, Chuanxia Zheng, Bohan Zhuang, Marc Pollefeys, Andreas Geiger, Tat-Jen Cham, and Jianfei Cai. MVSplat: Efficient 3d Gaussian splatting from sparse multi-view images. In *European Conference on Computer Vision (ECCV)*, 2024. 2, 3

[8] Siddharth Dalal et al. A review of recent advances in 3D Gaussian Splatting for optimization and reconstruction. *Image and Vision Computing*, 2024. 4

[9] Tianyu Ding et al. 3D Gaussian Splatting: Survey, technologies, challenges, and opportunities. *IEEE Transactions on Circuits and Systems for Video Technology*, 2025. 4

[10] Zhiwen Fan, Wenyan Cong, Kairun Wen, Kevin Wang, Jian Zhang, Xinghao Ding, Danfei Xu, Boris Ivanovic, Marco Pavone, Georgios Pavlakos, Zhangyang Wang, and Yue

Wang. InstantSplat: Sparse-view SfM-free Gaussian splatting in seconds, 2024. 3

[11] Ben Fei et al. 3D Gaussian Splatting as a new era: A survey. *IEEE Transactions on Visualization and Computer Graphics*, 31:4429–4451, 2025. 4

[12] Mehdi Hosseinzadeh, Shin-Fang Chng, Yi Xu, Simon Lucey, Ian Reid, and Ravi Garg. G3Splat: Geometrically consistent generalizable Gaussian Splatting. In *arXiv preprint*, 2025. arXiv:2512.17547. 4

[13] Mu Hu, Wei Yin, Chi Zhang, Zhipeng Cai, Xiaoxiao Long, Kaixuan Wang, Hao Chen, Gang Yu, Chunhua Shen, and Shaojie Shen. Metric3dv2: A versatile monocular geometric foundation model for zero-shot metric depth and surface normal estimation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(12):10579–10596, 2024. arXiv:2404.15506. 2, 3, 6

[14] Binbin Huang, Zehao Yu, Anpei Chen, Andreas Geiger, and Shenghua Gao. 2D Gaussian Splatting for geometrically accurate radiance fields. *ACM Transactions on Graphics (SIGGRAPH)*, 43(4), 2024. 8, 9

[15] Lihan Jiang, Yucheng Mao, Linning Xu, Tao Lu, Kerui Ren, Yichen Jin, Xudong Xu, Mulin Yu, Jiangmiao Pang, Feng Zhao, Dahua Lin, and Bo Dai. AnySplat: Feed-forward 3d gaussian splatting from unconstrained views. *ACM Transactions on Graphics*, 2025. SIGGRAPH Asia 2025; arXiv:2505.23716. 2, 3, 4

[16] Ying Jiang, Chang Yu, Tianyi Xie, Xuan Li, Yutao Feng, Huamin Wang, Minchen Li, Henry Lau, Feng Gao, Yin Yang, and Chenfanfu Jiang. VR-GS: A physical dynamics-aware interactive Gaussian splatting system in virtual reality. In *ACM SIGGRAPH*, 2024. arXiv:2401.16663. 2, 4

[17] Gyeongjin Kang, Jisang Yoo, Jihyeon Park, Seungtae Nam, Hyeonsoo Im, Sangheon Shin, Sangpil Kim, and Eunbyung Park. SelfSplat: Pose-free and 3d prior-free generalizable 3d Gaussian splatting. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 2, 3

[18] Bingxin Ke, Anton Obukhov, Shengyu Huang, Nando Metzger, Rodrigo Caye Daudt, and Konrad Schindler. Repurposing diffusion-based image generators for monocular depth estimation. In *CVPR*, 2024. arXiv:2312.02145. 2, 3

[19] Alex Kendall and Yarin Gal. What uncertainties do we need in bayesian deep learning for computer vision? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. 6

[20] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D Gaussian Splatting for real-time radiance field rendering. *ACM Transactions on Graphics (SIGGRAPH)*, 42(4):1–14, 2023. 1, 2, 4, 5, 8, 9

[21] Vincent Leroy, Yohann Cabon, and Jerome Revaud. Grounding image matching in 3d with MASt3R. In *European Conference on Computer Vision (ECCV)*, 2024. 2, 3

[22] Jiafei Li, Wenze Liu, Yunzhu Tang, Ran Li, Yilun Li, Hang Zhu, Lin Shao, Chenjia Yuan, and Yi Shan. ManipLLM: Embodied multimodal large language model for object-centric robotic manipulation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 4

[23] Haotong Lin, Sili Chen, Junhao Liew, Donny Y. Chen, Zhenyu Li, Guang Shi, Jiashi Feng, and Bingyi Kang. Depth anything 3: Recovering the visual space from any views, 2025. 2, 3

[24] Youtian Lin, Zuozhuo Dai, Siyu Zhu, and Yao Yao. Gaussian-Flow: 4d reconstruction with dynamic 3d gaussian particle. In *CVPR*, 2024. arXiv:2312.03431. 4

[25] Yifan Liu, Keyu Liu, Chenxin Yi, Bohan Zhu, Jiawei Zhang, Songcen Liu, Wei Cai, and Xiaoxiao Wang. MonoSplat: Generalizable 3D Gaussian Splatting from monocular depth foundation models. *arXiv preprint arXiv:2505.15185*, 2025. 4

[26] Guanxing Lu, Shiyi Zhang, Ziwei Wang, Changliu Liu, Jiwen Lu, and Yansong Tang. ManiGaussian: Dynamic Gaussian splatting for multi-task robotic manipulation. In *European Conference on Computer Vision (ECCV)*, 2024. 4

[27] Guanxing Lu, Shiyi Zhang, Ziwei Wang, Changliu Liu, Jiwen Lu, and Yansong Tang. ManiGaussian++: Hierarchical Gaussian world model for bimanual manipulation. In *arXiv preprint arXiv:2412.04972*, 2025. 4

[28] Tao Lu, Mulin Yu, Linning Xu, Yuanbo Xiangli, Limin Wang, Dahua Lin, and Bo Dai. Scaffold-GS: Structured 3D Gaussians for view-adaptive rendering. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. Highlight. 4, 8, 9

[29] Jonathon Luiten, Georgios Kopanas, Bastian Leibe, and Deva Ramanan. Dynamic 3d gaussians: Tracking by persistent dynamic view synthesis. In *3DV*, 2024. arXiv:2308.09713. 4

[30] (see arXiv 2604.02915). GP-4DGS: Probabilistic 4D Gaussian splatting from monocular video via variational Gaussian processes, 2026. 4

[31] Brandon Smart, Chuanxia Zheng, Iro Laina, and Victor Adrian Prisacariu. Splatt3R: Zero-shot Gaussian splatting from uncalibrated image pairs. In *arXiv preprint*, 2024. 3

[32] Fabio Tosi et al. NeRF: Neural radiance field in 3d vision: A comprehensive review. *arXiv preprint arXiv:2210.00379*, 2024. Updated post-Gaussian Splatting. 4

[33] Matias Turkulainen, Xuqian Ren, Iaroslav Melekhov, Otto Seiskari, Esa Rahtu, and Juho Kannala. DN-Splatter: Depth and normal priors for gaussian splatting and meshing. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2025. 2, 3

[34] Hengyi Wang and Lourdes Agapito. 3d reconstruction with spatial memory (Spann3R). In *International Conference on 3D Vision (3DV)*, 2025. 2, 4

[35] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. VGGT: Visual geometry grounded transformer. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. Best Paper Award. 2, 3, 4

[36] Ruicheng Wang, Sicheng Xu, Cassie Dai, Jianfeng Xiang, Yu Deng, Xin Tong, and Jiaolong Yang. MoGe: Unlocking accurate monocular geometry estimation for open-domain images with optimal training supervision. In *arXiv preprint*, 2024. 2, 3, 5

[37] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. DUSt3R: Geometric 3d

vision made easy. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2, 3

[38] Xihan Wang, Dianyi Yang, Yu Gao, Yufeng Yue, Yi Yang, and Mengyin Fu. GaussianGraph: 3d gaussian-based scene graph generation for open-world scene understanding. *arXiv preprint arXiv:2503.04034*, 2025. 4

[39] Yunsong Wang, Tianxin Huang, Hanlin Chen, and Gim Hee Lee. FreeSplat: Generalizable 3d Gaussian splatting towards free-view synthesis of indoor scenes. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 3

[40] Yi Wei, Shaohui Liu, Yongming Rao, Wang Zhao, Jiwen Lu, and Jie Zhou. NerfingMVS: Guided optimization of neural radiance fields for indoor multi-view stereo. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021. 4

[41] Guanjun Wu, Taoran Yi, Jiemin Fang, Lingxi Xie, Xiaopeng Zhang, Wei Wei, Wenyu Liu, Qi Tian, and Xinggang Wang. 4d gaussian splatting for real-time dynamic scene rendering. In *CVPR*, 2024. arXiv:2310.08528. 4

[42] Tong Wu, Yu-Jie Jia, Ling-Hao Li, et al. Recent advances in 3D Gaussian Splatting. *Computational Visual Media*, 2024. 4

[43] Tianyi Xie, Zeshun Zong, Yuxing Qiu, Xuan Li, Yutao Feng, Yin Yang, and Chenfanfu Jiang. PhysGaussian: Physics-integrated 3D gaussians for generative dynamics. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. Highlight. 2, 4

[44] Haofei Xu, Songyou Peng, Fangjinhua Wang, Hermann Blum, Daniel Barath, Andreas Geiger, and Marc Pollefeys. DepthSplat: Connecting Gaussian splatting and depth. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 2, 3

[45] Xinli Xu, Wenhang Ge, Dicong Qiu, ZhiFei Chen, Dongyu Yan, Zhuoyun Liu, Haoyu Zhao, Hanfeng Zhao, Shunsi Zhang, Junwei Liang, and Ying-Cong Chen. GaussianProperty: Integrating physical properties to 3d gaussians with LMMs. In *ICCV*, 2025. arXiv:2412.11258. 2, 4

[46] Jianing Yang, Alexander Sax, Kevin J. Liang, Mikael Henaff, Hao Tang, Ang Cao, Joyce Chai, Franziska Meier, and Matt Feiszli. Fast3R: Towards 3d reconstruction of 1000+ images in one forward pass. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 4

[47] Lihe Yang, Bingyi Kang, Zilong Huang, Zhen Zhao, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. Depth anything V2. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 2, 3

[48] Botao Ye, Sifei Liu, Haofei Xu, Xueting Li, Marc Pollefeys, Ming-Hsuan Yang, and Songyou Peng. No pose, no problem: Surprisingly simple 3d Gaussian splats from sparse unposed images. In *International Conference on Learning Representations (ICLR)*, 2025. 2, 3

[49] Zehao Yu, Anpei Chen, Binbin Huang, Torsten Sattler, and Andreas Geiger. Mip-Splatting: Alias-free 3D Gaussian Splatting. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. Best Student Paper. 8, 9

[50] Kai Zhang, Sai Bi, Hao Tan, Yuanbo Xiangli, Nanxuan Zhao, Kalyan Sunkavalli, and Zexiang Xu. GS-LRM: Large reconstruction model for 3d Gaussian splatting. In *European Conference on Computer Vision (ECCV)*, 2024. 2, 3

[51] Haoyu Zhen, Xiaowen Qiu, Peihao Chen, Jincheng Yang, Xin Yan, Yilun Du, Yining Hong, and Chuang Gan. 3D-VLA: A 3D vision-language-action generative world model. In *International Conference on Machine Learning (ICML)*, 2024. 2

[52] Haoran Zhou and Gim Hee Lee. MotionScale: Reconstructing appearance, geometry, and motion of dynamic scenes with scalable 4D Gaussian Splatting. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. arXiv:2603.29296. 4

[53] Shijie Zhou, Haoran Chang, Sicheng Jiang, Zhiwen Fan, Zehao Zhu, Dejia Xu, Pradyumna Chari, Suya You, Zhangyang Wang, and Achuta Kadambi. Feature 3DGS: Supercharging 3D Gaussian splatting to enable distilled feature fields. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 4

[54] Sicheng Zuo, Wenzhao Zheng, Yuanhui Huang, Jie Zhou, and Jiwen Lu. GaussianWorld: Gaussian world model for streaming 3d occupancy prediction. In *CVPR*, 2025. arXiv:2412.10373. 2, 4